Electro-optic waveguide TE ↔ TM mode converter with low drive voltage

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We have achieved complete electro-optic TE ↔ TM mode conversion in Ti-diffused lithium niobate waveguides with as little as a 2.5-V drive. Because, for birefringent substrates like lithium niobate, TE ↔ TM conversion is wavelength dependent, our method is potentially useful for wavelength multiplexing and demultiplexing. Three independently controlled mode converters, each with a 15-Å filter bandwidth and with center wavelengths separated by 80 Å, have been fabricated on a single substrate.

Recently we reported the demonstration of efficient phase-matched electro-optic TE ↔ TM mode conversion in Ti-diffused lithium niobate waveguides using periodic finger electrodes.[1] In this Letter we report the achievement of waveguide TE ↔ TM conversion with significantly reduced drive voltage by using interdigital electrodes. We have achieved essentially 100% conversion efficiency with as little as a 2.5-V drive. The mode conversion is strongly wavelength selective[1]; filter bandwidths between 5 and ~50 Å have been achieved by using device lengths of 6 to 0.5 mm, respectively. In addition, as a feasibility demonstration for wavelength demultiplexing applications, we have integrated on a single substrate three mode-converter filters, each with a ~15-Å filter bandwidth and center wavelengths separated by ~80 Å.

We achieved efficient, low-voltage electro-optic polarization conversion in lithium niobate by using the strong[2] r_{51} ($=28 \times 10^{-12}$ m/V) coefficient. Utilization of this coefficient requires periodic electrodes for phase matching the TE and TM modes, which propagate with different velocities because of the large material birefringence of lithium niobate. In our previous work, a Z-cut crystal with X-directed electric fields using finger electrodes was employed.[1] This orientation has the advantage of not requiring electrodes to be placed over the waveguides; thus no intermediate dielectric buffer layer is needed.[3] However, as described in this Letter, we have significantly reduced the required drive voltage for the same device length by using interdigital electrodes.

A schematic drawing and photomicrograph of a section of a typical device illustrating the interdigital electrodes are shown in Fig. 1. The r_{51} coefficient is employed by using an X-cut, Y-propagating crystal with X-directed fields. The TE ↔ TM coupling coefficient, κ, is[4,5]

$$\kappa = \alpha \frac{\pi}{\lambda} n_s{}^3 r_{51} \frac{V}{d}, \tag{1}$$

where V is the applied voltage, d is the interelectrode gap, n_s is the substrate's refractive index, λ is the optical wavelength, and α is a parameter between 0 and 1, which is a measure of the overlap between the applied electric field and the TE–TM modal fields. For fixed parameters this expression is a factor of 2 larger than that for finger electrodes[1] because of the push–pull effect of the interdigital electrodes [see Fig. 1(a) inset].

For a given electrode period Λ, complete conversion efficiency is achievable only for λ_0 such that the phase-match condition[6]

$$\frac{2\pi}{\lambda_0} |N_{TE}-N_{TM}| = \frac{2\pi}{\Lambda} \tag{2}$$

is satisfied. N_{TE} and N_{TM} are the effective indices for the TE and TM modes, respectively. Finite mismatch for $\lambda \neq \lambda_0$ results in reduced conversion efficiency and thus in the wavelength selectivity. Polarization conversion is characterized analytically by the codirectional coupled-wave equations. The expected passband-filter response is given by the solution to these equations.[1,7] The normalized spectral bandwidth (FWHM) is $\Delta\lambda/\lambda_0 = \Lambda/L = 1/N$, where L is the number of electrode periods.

Waveguides that support a single TE and TM mode were fabricated by indiffusing a 2-μm-wide, ~290-Å-thick strip of titanium into an X-cut, Y-propagating lithium niobate crystal. The diffusion was performed

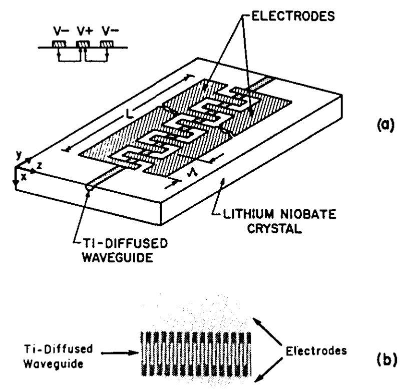

Fig. 1. (a) Schematic drawing and (b) photomicrograph of electro-optic waveguide TE ↔ TM mode converter with periodic interdigital electrodes.

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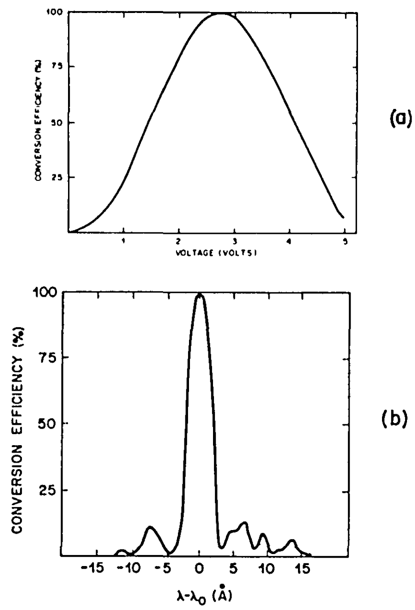


Fig. 2. (a) Measured phase-matched ($\lambda = \lambda_0 \sim 0.6\ \mu$m) TE $\rightarrow$ TM conversion efficiency versus drive voltage and (b) measured wavelength response for fixed voltage ($V \sim 2.5$ V) for electrode length of 6 mm.

at 980°C for 4 h in a flowing argon atmosphere with cooldown in flowing oxygen. Both gases were bubbled through water at a flow rate of ~1 liter/min. After diffusion, a ~1200-Å-thick SiO_2 buffer layer was deposited over the waveguide by chemical-vapor deposition[10] to eliminate the high loss to the TM mode caused by the overlaid metal electrodes.[3] An electron-beam-written mask was used to define chrome–aluminum interdigital electrodes photolithographically over the waveguide, as shown in Fig. 1. The electrode period is $\Lambda = 7\ \mu$m, which requires the delineation of 1.75-μm-wide electrodes (Fig. 1). The crystal ends were polished to allow end-fire coupling. Devices with various electrode lengths were fabricated.

In spite of the X-cut crystal orientation, we have repeatedly fabricated strip waveguides without the undesirable planar guiding that others have reported as being due to an increase in the extraordinary index from lithium oxide outdiffusion.[8] No special techniques, other than the described diffusion process with flow gases bubbled through water, were used to eliminate or compensate for outdiffusion.[9] Using the same diffusion process, we have also repeatedly fabricated strip waveguides in Z- and Y-cut lithium niobate that are free of planar guiding for both polarizations.[1,10,11]

The measured TE $\leftrightarrow$ TM conversion efficiency (using output intensities) for phase-matched coupling versus applied voltage ($L = 6$ mm) is shown in Fig. 2(a). A peak conversion efficiency greater than 99% was achieved with an applied voltage of ~2.5 V. This represents a fivefold voltage reduction compared with a device of the same length with finger electrodes.[1] The reduction is achieved in spite of the need for the SiO_2 buffer layer, which generally necessitates a larger applied voltage to achieve the same electric-field strength within the crystal as compared with the nonbuffered case. The voltage reduction here results from the push–pull effect of the interdigital electrodes, a smaller interelectrode gap, d (~1.75 μm compared with 5 μm for the finger electrodes of Ref. 1), and possibly a better overlap in the electric and modal fields for this geometry.

It is important to note that, for interdigital electrodes, the value of d is directly related to the required period, $d \simeq \Lambda/4$. Therefore, these electrodes are especially preferable for use with highly birefringent crystals, such as lithium niobate, in which Λ is small. On the other hand, the value of d for finger electrodes can be chosen independently of the period, with only the limitation that d be larger than the waveguide width. Consequently, finger electrodes may actually permit lower drive voltages for low birefringent crystals in which large-period electrodes are required.

The measured mode-conversion wavelength response of this device is shown in Fig. 2(b). The center wavelength is $\lambda_0 \sim 0.600\ \mu$m, and the bandwidth (FWHM) is ~5 Å. Experimental verification of the expected filter bandwidth dependence on electrode length is shown in Fig. 3. As indicated, we have achieved bandwidths between 5 and ~50 Å with device lengths from 6 to 0.5 mm, respectively. Because of the strong coupling coefficient for this electrode geometry, complete mode conversion for the 500-μm-long device was achieved with a drive voltage of only ~20 V. The measured results—including λ_0 and the drive voltage required for complete mode conversion—were identical for TE $\rightarrow$ TM and TM $\rightarrow$ TE conversion, as expected.

As a feasibility demonstration for wavelength multiplexing–demultiplexing applications, we have fabricated three mode converters with slightly different electrode periods placed sequentially over a single waveguide. Voltage can be independently applied to each device. The desired small electrode-period differences could not be achieved with the 0.25-μm address quantization of the electron-beam-exposure system used for mask generation. Instead it was achieved by tilting a 7-μm-period pattern at angles of 0°, 10.5°, and 14.8°, as shown in Fig. 4(b). The resulting electrode

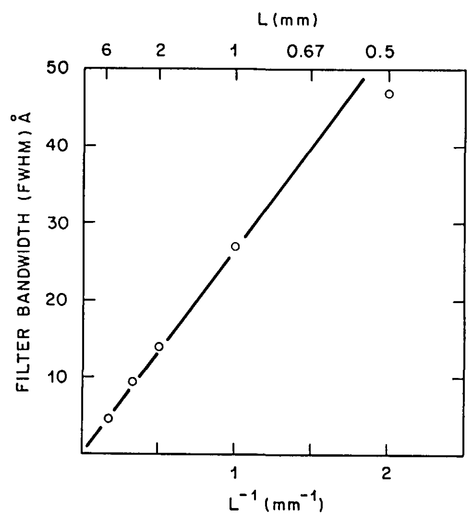
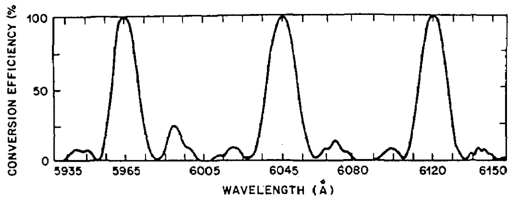
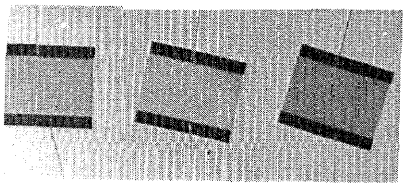


Fig. 3. Measured mode-conversion filter bandwidth versus electrode length.



(a)



(b)

Fig. 4. (a) Measured TE → TM conversion efficiency filter response for three independently controlled mode converters with periods $\Lambda = 7, 7.12$, and 7.24 μm placed along a single waveguide and (b) photograph of the device.

periods are 7, 7.12, and 7.24 μm, respectively. Each device is ~2 mm long. The measured wavelength-selective TE ↔ TM conversion efficiency obtained by sequentially applying voltage to each device is shown in Fig. 4(a). Essentially complete polarization conversion was achieved for center wavelengths separated by ~80 Å. The bandwidth of each device is ~15 Å. The measured cross talk at the center wavelength of one device with voltage applied to an adjacent device was ~−20 dB. This cross-talk result is in good agreement with theory[12] for the ratio of wavelength separation to filter bandwidth used.[12] For fixed filter bandwidth, the center wavelength spacing can be reduced without increasing cross talk by carefully weighting the coupling coefficient to reduce filter sidelobes.[12] Of course, for demultiplexing applications, a waveguide polarization splitter[13] will be required after each mode converter.

No accurate transmission-loss measurements were made. However, measurements reported for similarly prepared waveguides indicate ~0.5-dB/cm larger loss for waveguides loaded with Al electrodes over an intermediate dielectric buffer layer compared with unloaded ones (~1.5 dB/cm compared with ~1 dB/cm).[14] Thus, for typical device lengths, the extra loss for interdigital electrodes is minimal.

In summary, by using interdigital electrodes, we have demonstrated efficient, low-voltage waveguide electro-optic TE ↔ TM conversion in a Ti-diffused lithium niobate waveguide. The wavelength selectivity of this interaction has been examined. Filter bandwidths of 5 to ~50 Å have been demonstrated. Three mode converters with center wavelengths separated by 80 Å each with 15-Å bandwidths have been integrated on a single lithium niobate substrate.

The photomask computer program was written by R. Bosworth.

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